
How Foveated Vision Shapes Cognitive Maps in Navigation Agents

Anonymous authors

Paper under double-blind review

Abstract

Spatial memory in primates is not a sensor-independent module. Hippocampal cells code joint place–view–action state in part because the primate eye samples the world through a foveated sensor; rodents with near-panoramic vision instead develop classic place fields, and individuals deprived of vision recruit hippocampal circuits to compensate. The biological pattern is consistent: *the structure of the sensor shapes the structure of spatial memory*. Whether the same holds for artificial navigation agents — whose learned memory also develops cognitive-map-like representations under task pressure alone — has not been tested directly, because prior work has either removed vision entirely or used spatially uniform input. We isolate the principle in a controlled silicon case study: five recurrent (LSTM) PointNav agents trained identically except for the visual pathway, spanning a sensor-structure spectrum from no vision, through a resolution-collapsed encoder, foveated vision (fixed and learned gaze), to full uniform vision. Probing their recurrent state reveals an *encoder–memory race*: the linearly-decodable spatial code is maintained when the encoder supplies little spatial-feature variety per step, and dissipates when the encoder is rich enough to substitute. Independently, agents at comparable task competence develop hidden states in linearly disjoint subspaces, and cross-condition memory transplants are asymmetric: bottleneck-donor states hurt rich-encoder recipients more than vice versa. The intuition that “richer perception begets richer memory” fails in this regime — the relationship across conditions is inverse — making encoder capacity a candidate training-time lever for cognitive-map-like representations (single-seed; multi-seed replication and a causal scaling sweep are in flight). The pattern echoes biological findings on hippocampal compensation and cross-modality remapping, suggesting a sensor–memory coupling principle whose generality across architectures is open.

1 Introduction

Primate spatial memory is shaped by how primates see. Hippocampal cells code joint place–view–action state in part because the primate eye samples the scene through a foveated sensor [Wirth et al., 2017, Rolls, 2024]; rodents with near-panoramic vision instead show classic place fields. Sensor structure also reorganises memory *within* a single animal — bats produce non-overlapping hippocampal populations under vision vs. echolocation [Geva-Sagiv et al., 2016], and congenitally blind humans develop enhanced hippocampal coupling on non-visual spatial tasks [Kupers and Ptito, 2014]. The biological pattern is consistent: *the structure of the sensor reshapes the structure of spatial memory*. We ask whether the same is true of artificial navigation agents.

The emergent-cognitive-maps literature has shown that map-like structure arises in the learned internal state of navigation agents trained with deep RL, including agents with no vision at all [Wijmans et al.,

2023, Banino et al., 2018, Cueva and Wei, 2018].¹ These results establish that *some* map-like structure is learned without being designed in. They leave open the complementary question we take up here: holding task and architecture fixed, does the structure of the visual sensor reshape *what* the agent’s memory encodes? We isolate sensor structure as the only varied component across five conditions spanning a sensor-structure spectrum (Figure 1): no vision, a resolution-collapsed encoder, foveated vision with fixed gaze, foveated vision with learned gaze, and uniform full-resolution vision. The resolution-collapsed control distinguishes outright blindness from the case where vision is present but the encoder cannot spatially resolve it; foveation occupies a biologically motivated point in between. *We use a recurrent (LSTM-based) PointNav agent as our case study*: this matches the architecture of the emergent-maps literature we extend [Wijmans et al., 2023], gives every cross-condition comparison a single hidden-state object to probe, and provides a tractable controlled setting. The principle we test — whether sensor structure shapes the format of learned memory — is stated in architecture-agnostic terms; whether it transfers to transformer-based or supervised visual learners is a scope question we discuss in §5, not establish here.

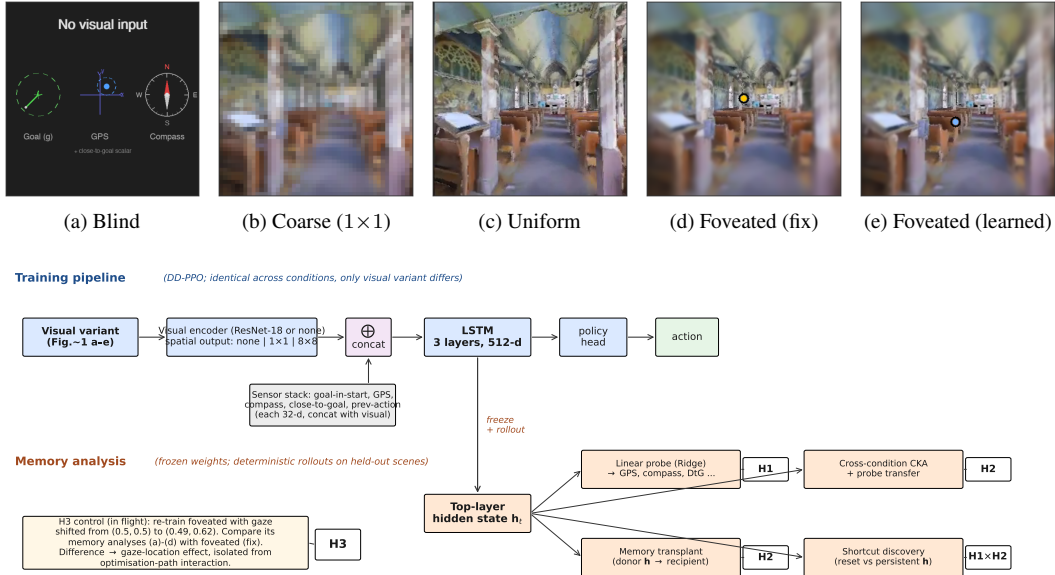


Figure 1: **Setup.** *Top row, (a)–(e)*: the five visual sensor conditions, shown on the same scene. (a) *Blind*: no visual stream; only the non-visual sensor stack. (b) *Coarse (1×1)*: 48×48 uniform RGB, chosen so the ResNet-18 encoder collapses spatially to a 1×1 feature map. (c) *Uniform*: 256×256 full-resolution RGB. (d) *Foveated (fix)*: eccentricity-dependent Gaussian blur ($\sigma_{\max} = 8$, quadratic falloff) centred at the image middle (yellow dot). (e) *Foveated (learned)*: same blur model with a learned (x, y) gaze offset (blue dot, post-training collapsed location (0.49, 0.62)). *Bottom block*: the structural variable being manipulated is encoder spatial output (none | 1×1 | 8×8); the DD-PPO learner, sensor stack, LSTM backbone, and policy head are otherwise identical. After training, weights are frozen and we collect deterministic rollouts to extract top-layer hidden states h_t , on which we run linear probes (H1 evidence), cross-condition CKA + probe transfer (H2 evidence), memory transplants (H2 behavioural validation), and shortcut discovery (probe-readable vs. policy-used dissociation, §4.5).

We test three hypotheses. *H1 (encoder–memory race)*: when the visual encoder supplies little spatial-feature variety per step, the recurrent state should compensate by carrying a world-frame spatial code; richer encoder output should let memory rely on visual features instead. *H2 (format divergence)*: different sensor conditions should produce hidden-state representations in distinct linear subspaces, not scaled versions of a common code. *H3 (gaze location)*: within the rich-encoder regime, where a

¹We use “cognitive map” loosely throughout for a linearly-decodable world-frame position code in the agent’s hidden state — the operational quantity our probes measure. This is narrower than the cognitive-map concept in the broader neural-and-cognitive-science literature (allocentric, relational, multi-scale, hierarchical maps; see e.g. [Epstein et al., 2017]). Our claims should be read as scoped to the linearly-decodable subset; richer non-linear or relational structure may exist in the same hidden states and is not addressed by linear probes.

foveated sensor is centred should act as a second content axis independent of the sensor transform. Each has a direct biological analog (§5.2).

Three findings emerge with the strongest evidence. An encoder–memory race along the sensor-structure spectrum (H1) that lives inside the recurrent state, not at the encoder. The five agents solve the task at comparable competence but develop hidden states in linearly disjoint subspaces, with measurable cross-condition memory-transplant cost (H2). And probe-readable vs. policy-used memory dissociates in opposite directions for two specific conditions, suggesting that “has a cognitive map” and “uses a cognitive map” need not coincide. Foveation under our blur model converges with uniform on H1 but diverges on H2, behaviour, and dataset shift. H3 (gaze location) is tested by a foveated-shifted control in training at submission.

The cross-condition pattern is correlation-level; a scaling-sweep experiment that would test the H1 mechanism causally is in progress (Appendix H). Foveation-specific predictions involving alternative foveation models depend on experiments still in training (§4.4, Appendix G). Each finding has a direct biological precedent which we discuss as convergent (not conclusive) evidence in §5.2.

2 Related Work

Cognitive maps in RL navigation agents Wijmans et al. [Wijmans et al., 2023] demonstrated that map-like representations emerge in the recurrent memories of DD-PPO agents even without visual input; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related spatial concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] found grid-cell-like units in recurrent networks trained on path integration; Gornet & Thomson [Gornet and Thomson, 2024] extended this via visual predictive coding. Hennig et al. [Hennig et al., 2023] showed that RL under partial observability can induce belief-state-like representations without explicit probabilistic objectives. Ramakrishnan et al. [Ramakrishnan et al., 2025] examined spatial cognition in frontier models. All these studies use either absent or spatially uniform perception; none vary input quality spatially, and none test whether different sensors produce different representational formats.

Foveated vision and gaze policies in deep learning Foveated rendering [Strasburger et al., 2011, Rosenholtz, 2016] exploits eccentricity-dependent acuity falloff for efficiency. Learned glimpse policies appear in recurrent attention models [Mnih et al., 2014]; Atanov et al. [Atanov et al., 2024] showed that sensor layout changes downstream task performance. Pourrahimi & Bashivan [Pourrahimi and Bashivan, 2025] probed a foveated visual-search model for neuroscience-predicted features; Shang & Ryoo [Shang and Ryoo, 2023] jointly learned gaze and motor policies following the Sprague & Ballard [Sprague and Ballard, 2003] framework. These address visual search or task performance; none asks how sensor structure reshapes the learned spatial memory of a downstream navigation policy.

Probing neural representations Belinkov [Belinkov, 2022] surveys probing classifiers; Hewitt & Liang [Hewitt and Liang, 2019] introduced control tasks to distinguish genuine structure from probe memorisation; Kornblith et al. [Kornblith et al., 2019] introduced linear CKA for cross-model similarity. We adopt linear probing, the Hewitt–Liang selectivity metric, and unaligned linear CKA, and complement them with two behavioural interventions directly on the recurrent state (memory transplant and shortcut discovery) to validate the probe-based findings outside the linear-probe framework.

What is new here Three things distinguish this work. (i) A *controlled* five-condition ablation with everything but the visual pathway held identical, including the coarse condition that decouples “no vision” from “vision the encoder cannot resolve”. (ii) A protocol-corrected probing methodology (deterministic action selection at probe time; stochastic sampling inflates probe R^2 via a 4-step quasi-static-trajectory artefact, Appendix A). (iii) Convergent behavioural validation via memory transplant and shortcut discovery, outside the linear-probe framework. Biological motivation and the parallel to animal-navigation research are integrated in §5.2 rather than recapitulated here.

3 Methods

3.1 Setup

All agents are trained on PointGoal navigation in Habitat [Savva et al., 2019] using DD-PPO [Wijmans et al., 2020] on a merged pool of Gibson-0+ [Xia et al., 2018] (411 scenes) and Matterport3D train (61 scenes), totalling 472 training scenes; evaluation uses 18 held-out MP3D test scenes (1008 episodes). All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar), each projected to 32-d and concatenated with a 32-d previous-action embedding. All sighted conditions train to $\sim 250\text{M}$ environment frames; the blind baseline trains to 342M to allow slower convergence from proprioception alone.²

The five visual conditions (Figure 1): *blind* (no visual input); *uniform* (full 256×256 RGB encoded by ResNet-18); *foveated (fix)* (eccentricity-dependent Gaussian blur, $\sigma_{\max} = 8$ quadratic falloff, gaze locked at image centre, same ResNet-18 encoder)³; *coarse* (48×48 uniform RGB; at this resolution ResNet-18’s 32-fold spatial downsampling collapses the feature map to 1×1 , removing spatial structure but preserving channel information); *foveated (learned)* (same blur model, gaze centre predicted by a lightweight MLP trained end-to-end). Coarse is the spatial-bandwidth lower bound for our encoder; it has visual input but its encoder output cannot resolve world-frame position. The 48×48 input also has lower spatial resolution than uniform’s 256×256 , so this single condition cannot fully separate the 1×1 collapse from low input resolution; the encoder-resolution scaling sweep (Appendix H) addresses this directly. The foveated conditions disable the running-mean-and-variance input normaliser to avoid an in-place autograd conflict along the gaze-decoder gradient path; an invariance control is in training. We also discovered and patched a silent NaN-gradient corruption mode in DD-PPO (Appendix F).

3.2 Probes

After training, all weights are frozen. We collect per-step top-layer LSTM hidden states (512-d) paired with ground-truth labels from $n=500$ Gibson held-out episodes under *deterministic* argmax-action rollouts (matching our shortcut / transplant / standard eval protocols; the alternative stochastic protocol gives trivial ~ 4 -step rollouts and is documented in Appendix A).⁴ We train Ridge probes ($\alpha = 10$) with episode-level 5-fold cross-validation, report mean $\pm \sigma$ across folds, and verify probe specificity with Hewitt–Liang label-permutation selectivity [Hewitt and Liang, 2019]. Probed variables: GPS position, compass heading, distance-to-goal, lag- k path history ($k \in \{0, \dots, 20\}$), visited-region occupancy, full occupancy grid, per-unit spatial information, ego-frame goal vector. For H2 representational geometry we additionally use linear CKA [Kornblith et al., 2019] and cross-condition probe transfer.

We complement probes with two interventions on the LSTM state. *Memory transplant*: paste a donor’s hidden state into the recipient at step 30 and compare baseline / self-transplant / cross-transplant SPL

²Algorithm-and-task scope: DD-PPO + PointGoal in Habitat with ResNet-18 visual encoder. The H1/H2 patterns we report could in principle shift under different RL algorithms (e.g., on-policy A2C, IMPALA, off-policy SAC), different navigation tasks (ObjectGoal, ImageGoal, exploration), different simulators (iGibson, AI2-THOR, real-world), or different visual encoders (smaller CNN, ViT). Each is a deliberate scope choice for direct comparability with [Wijmans et al., 2023]; verifying generality is open.

³Foveation-model choice: Gaussian blur with quadratic eccentricity falloff is a tractable approximation, not a faithful model of biological retinal sampling. Real foveation involves spatially non-uniform sampling (log-polar or multi-scale pyramid) which directly reduces the encoder’s spatial output dimensionality, whereas Gaussian blur preserves it. The F3 log-polar control in training (Appendix G) is a falsifiable test of whether the encoder spatial-output story explains foveation under a different operationalisation; if log-polar matches uniform on H1 (rather than the predicted intermediate value), the H1 mechanism story would need reframing to depend on something other than encoder spatial output.

⁴We probe top-layer hidden state $\mathbf{h}_2$ in the main results because it is what the (linear) policy head reads. Linear-probe results across all 6 (state $\times$ layer) representations — $\mathbf{h}_0, \mathbf{h}_1, \mathbf{h}_2$ and $\mathbf{c}_0, \mathbf{c}_1, \mathbf{c}_2$ — are reported in Appendix Figure 9: cell states do *not* carry the GPS code as cleanly as hidden states, so the H1 “linearly-decodable top-layer GPS” claim is specifically about $\mathbf{h}_2$, not the cell-state pathway. The full per-condition CKA / transplant / shortcut machinery on $\mathbf{c}_t$ and $\mathbf{h}_0, \mathbf{h}_1$ is left for follow-up; if condition-specific format-divergence structure lives in $\mathbf{c}_t$ but not $\mathbf{h}_2$, the H2 “disjoint subspaces” claim would extend or weaken accordingly.

across 150 episodes per pair.⁵ *Shortcut discovery*: run paired same-scene-different-goal episodes (20 scenes $\times$ 10 pairs) and compare second-episode SPL under reset vs. persistent LSTM. For foveated (fix), all analyses use ckpt.36 at $\sim$ 174M frames (last clean intermediate before the silent-corruption window).

4 Results

4.1 Five conditions, same task

Table 1 collects the headline per-condition numbers. *All five agents solve the same task at near-equivalent competence*. Sighted conditions reach 96–99% success; blind reaches 93% at a longer training budget, reproducing Wijmans et al. [Wijmans et al., 2023]. The narrow success bracket rules out a skill-gap account for any encoded-content differences below. The probe machinery is calibrated: distance-to-goal decodes everywhere ($R^2 \in [0.81, 0.90]$), demonstrating Ridge has the capacity to fit high- R^2 scalars when the hidden state contains them; Hewitt–Liang label-permutation selectivity is high precisely where GPS is high (Appendix A). The behavioural near-equivalence sets up the question the rest of §4 addresses: *what differs internally*, given that what the agents *do* is the same?

Table 1: Per-condition headline. Probe R^2 columns are the 5-fold episode-level CV mean $\pm$ std on full-length deterministic-rollout trajectories (no per-episode step cap). *GPS* and *Compass* are world-frame. The bottleneck-vs-rich-encoder gap is in temporal *stability* of the code, not its mid-episode existence (Figure 2b): all five conditions decode top-layer GPS at $R^2 \in [0.6, 0.95]$ in the typical-episode window (steps 50–200); the rich-encoder long-tail collapse is what produces the sub-chance numbers below. Bold marks the H1-relevant ordering (blind $>$ coarse $\gg$ rich-encoder).

Condition	Frames	SPL	Success	GPS R^2 (no-cap)	Compass R^2 (no-cap)
Blind	342M	0.59	0.93	+0.95$\pm$0.02	+0.81$\pm$0.08
Coarse	250M	0.67	0.98	+0.78$\pm$0.10	+0.64$\pm$0.10
Uniform	250M	0.85	0.99	−0.31 $\pm$ 0.86	+0.36 $\pm$ 0.23
Foveated (fix)	174M	0.78	0.96	+0.06 $\pm$ 0.88	+0.07 $\pm$ 0.69
Foveated (learned)	250M	0.82	0.98	−2.43 $\pm$ 3.98	−1.34 $\pm$ 3.14

4.2 Encoder–memory race: a temporal-stability claim (H1)

The headline finding is non-obvious: visual-encoder capacity tracks top-layer LSTM spatial encoding *inversely*. Conditions with the smallest visual-feature throughput per step (blind, coarse) carry a strong linear GPS code at the policy readout, and conditions with the richest visual stream (uniform, foveated, foveated (learned)) do not (Figure 2a). Goal-distance, included as a sanity check, decodes everywhere.⁶

Temporal stability is the precise claim Step-binned probing (Figure 2b) separates “never had a top-layer GPS code” from “had one mid-episode then lost linear readability”; rich-encoder conditions are the latter. The Table 1 “rich-encoder $\approx$ chance” headline is therefore long-tail behaviour, not mid-episode absence. *H1 is most precisely a temporal-stability claim about the linearly-decodable top-layer GPS code*. A lag- k probe corroborates: blind sustains GPS at $k = 20$, while rich-encoder conditions are at chance for every $k \leq 10$.

Pipeline view: where GPS sits, and where it disappears. Panel (c) of Figure 2 traces GPS R^2 across four locations along the agent’s information path. Three readings come out together. (i) *The visual encoder does not linearly carry GPS*: encoder feature-map probes for coarse, uniform, and foveated all sit at or below chance ($R^2 = -3.14, -0.32, -0.65$ respectively). The H1 divergence is therefore not an encoder-level position-retention property — the encoders are roughly equally GPS-blind. (ii)

⁵Transplant midpoint $t=30$ is a single design choice. The supplementary midpoint sweep across $\{0, 15, 30, 60, 90, 200, 400, 800\}$ steps (Figure 8) shows the cross-self gap peaks in the $t \in [30, 90]$ window (where mismatch most matters — the agent has integrated some history but is not yet committed) and decays toward 0 at late midpoints ($t \geq 400$, where the policy is already committed to its trajectory). Earlier midpoints ($t < 15$) collapse to a protocol-noise floor.

⁶Whether the correlation reflects causation is what the §4.4 experiments and the Appendix H resolution sweep test directly. All numerical claims in §4.2 are single-seed; multi-seed replication of all five conditions is in flight.

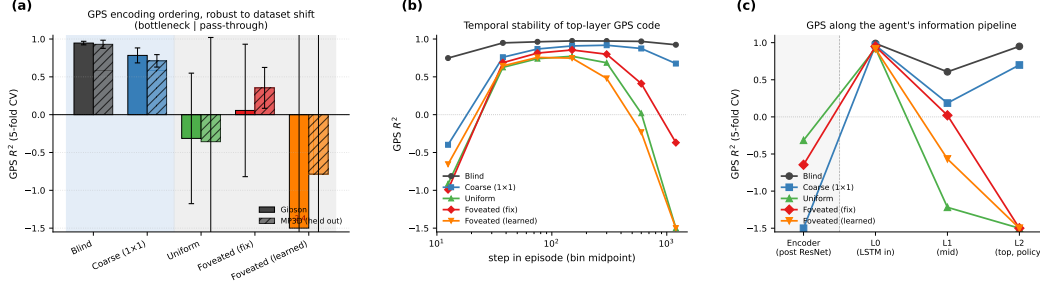


Figure 2: H1 evidence in three panels. (a) Top-layer GPS R^2 (5-fold CV, deterministic rollouts) on Gibson and held-out MP3D (300 episodes/condition). Blind and coarse (bottleneck) decode world-frame GPS on both datasets; uniform / foveated / foveated (learned) (rich-encoder) do not, except foveated recovering on MP3D. (b) Top-layer GPS R^2 binned by step-in-episode: all five conditions encode top-layer GPS in the typical window (steps ~ 25 –200); the code becomes no longer linearly readable in the rich-encoder long-tail (negative R^2 in the 800+ step bin), while bottleneck conditions hold $R^2 > 0.7$ to step 800+. The headline in (a) is therefore primarily long-tail behaviour. (c) GPS R^2 along the agent’s information pipeline: Encoder (post-ResNet-18 feature-map probe, pre-LSTM) $\rightarrow$ L0 (LSTM input layer, where the sensor stack concatenates with the visual encoder output) $\rightarrow$ L1 (middle) $\rightarrow$ L2 (top, policy-readable). For sighted conditions, the encoder is at chance/negative (no sighted encoder linearly carries GPS); the L0 jump is the GPS-sensor concat. The H1 divergence is at L2: bottleneck (blind, coarse) hold; rich-encoder discard. Blind has no encoder; foveated (learned) lacks the encoder probe but follows the L0–L2 pattern.

The L0 GPS comes from the sensor concat, not the network “computing” position: every condition sees a sharp jump from encoder to L0 ($R^2 \rightarrow \sim 0.95$) because the GPS sensor embedding enters the LSTM input vector at L0. (iii) The H1 divergence is specifically at the policy readout (L2): bottleneck conditions hold their L0 GPS through L2; rich-encoder conditions discard it. Since the DD-PPO action head is a linear projection of h_2 , this maps directly to “policy-accessible”. A 2-layer MLP on the same L2 states recovers GPS for rich-encoder conditions ($R^2 \in [0.51, 0.73]$), so H1 is precisely about linear L2 decodability.⁷ The blind result ($R^2 = 0.95$) replicates Wijmans et al. [Wijmans et al., 2023]; coarse extends their finding to a condition that has visual input but a 1×1 encoder output, so the shared trigger is minimal visual-feature variety per step, not absence of vision.

Proposed mechanism We propose H1 is a substitution effect: integrating the Layer-0 GPS-sensor input across time and exploiting the encoder’s position-correlated visual features are alternative routes to the same policy output. Bottleneck conditions force the integration route (no rich visual features available), so the temporally-stable GPS code persists; rich-encoder conditions provide a viable visual route, so the integrated code is no longer needed by the policy and ceases to be linearly readable. The early-then-decay temporal profile in Figure 2b is the direct fingerprint this account predicts.⁸

4.3 Format-level divergence across conditions (H2)

Behavioural test (memory transplant) Pasting a donor’s mid-episode hidden state into a recipient’s policy at step 30 tests whether the representational geometry differences *matter* outside the probe framework (Figure 3, right). Three structural facts emerge that no linear-similarity measure on its own would surface. *Asymmetry along the bottleneck axis*: bottleneck-donor states are toxic to rich-encoder recipients while the reverse direction is benign — the rich-encoder LSTM cannot make sense of bottleneck-donor activity at the policy readout, but a bottleneck LSTM can still drive its policy from rich-encoder donor activity. *Recipient brittleness scales with encoder bandwidth*: uniform suffers the largest cost from *any* foreign donor; blind tolerates them all. *The linear-similarity blind spot*: pairs

⁷Two single-seed cross-dataset shifts on MP3D — foveated GPS at chance $\rightarrow +0.35$, foveated (learned) compass $-1.34 \rightarrow +0.41$ — await multi-seed replication; if they survive, they would suggest condition-specific MP3D recovery worth investigating, but at $N = 1$ we cannot rule out per-seed noise. The MLP result alone cannot exclude exploitation of position-correlated features; Appendix B discusses both.

⁸A direct causal test — ablating the visual stream mid-rollout in rich-encoder agents to see whether the top-layer GPS code re-emerges, or perturbing the GPS sensor input mid-rollout in bottleneck agents to see whether SPL collapses — is left for future work.

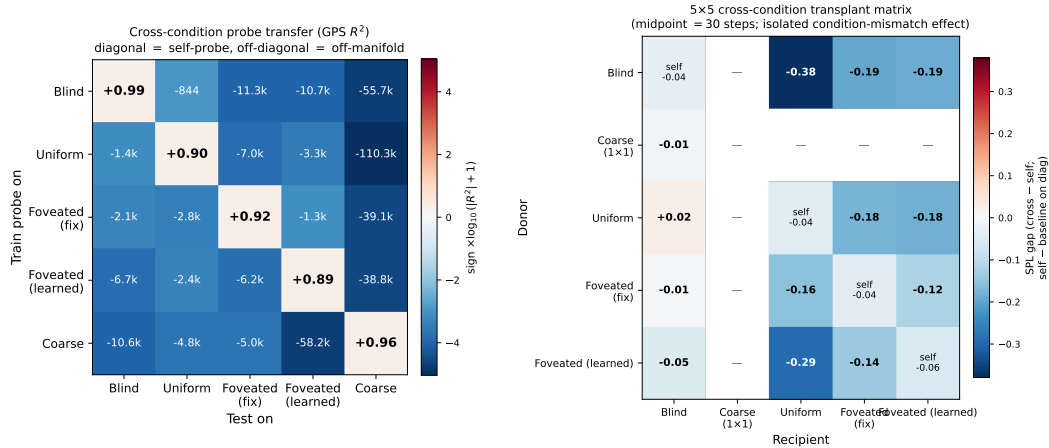


Figure 3: H2 convergent evidence, two views. *Left*: cross-condition probe transfer for GPS R^2 (train probe on row, test on column; signed-log colour scale). The diagonal is +0.89–+0.99 (self-probe), every off-diagonal predicts off-manifold ($R^2 \ll -800$, often $\ll -10^4$). *Right*: 5x5 cross-condition memory-transplant matrix at midpoint $t=30$. Cells off-diagonal: cross-transplant SPL minus recipient’s self-transplant SPL (isolated condition-mismatch effect). Cells with “—” (coarse as recipient): not yet measured. Both heatmaps are asymmetric: bottleneck-donor states hurt rich-encoder recipients far more than the reverse, and probes trained on bottleneck states fail catastrophically on rich-encoder hidden states. A midpoint-sweep companion across $\{0, 15, 30, 60, 90\}$ is in Appendix D.

that read as indistinguishable under unaligned linear CKA differ by 0.01–0.38 SPL under transplant — “these representations look identical to a linear similarity metric” is consistent with “their actual policy-relevant content is incompatible”.⁹

Convergent evidence at the probe level. Cross-condition probe transfer (Figure 3, left) shows the same asymmetric pattern: probes trained on bottleneck states predict off-manifold by a wider margin on rich-encoder hidden states than the reverse. The 1-NN purity is 1.000 vs. chance 0.20 on 1500×5 pooled top-layer states — a sample’s nearest neighbour by Euclidean distance is always within its own condition. The probe-transfer failure is also *symmetrically catastrophic* (every off-diagonal cell), ruling out an embedding asymmetry where one condition’s code simply contains another’s.¹⁰

4.4 Foveation: convergence on H1, divergence on H2 + behaviour

H1 (§4.2) groups foveated (fix) with uniform as “rich-encoder pass-through” — a coarse partition. Within that group, every measure beyond top-layer linear GPS separates them (Table 2).

The H1 phenomenon (§4.2) classifies foveated (fix) and uniform together as rich-encoder pass-through. But every other measure separates them (Table 2): *whatever foveation does differently from uniform happens inside the LSTM in subspaces a linear top-layer GPS probe is blind to*. Compass at the top layer, MP3D held-out generalisation, shortcut behavioural reliance, and cross-condition transplant compatibility all read foveation $\neq$ uniform; pairwise CKA and 1-NN purity say their hidden states are linearly separable. The four in-flight foveation experiments (Appendix G) test *what* the differential is; F3 log-polar in particular makes a falsifiable prediction: a $\sim 2 \times 2$ encoder spatial output should produce LSTM GPS $R^2 \geq 0.3$, between coarse’s +0.78 and uniform’s ≈ 0 . If instead

⁹The coarse-recipient column is currently unmeasured (4 cells queued); the recipient-ranking statement uses non-coarse-recipient cells only. Single-seed; the asymmetry pattern is robust to landed cells but the magnitudes will move with replication.

¹⁰Methodological reassurance: probe capacity is not the bottleneck (self-probes succeed everywhere); sample-size asymmetry is ruled out by common-sample truncation; task competence is bunched (96–99%, §4.1), so the divergence is not a skill difference. Boundaries: similarity tests are linear or near-linear, so non-linear measures could reveal higher-order shared structure; the five sensors are diverse but not exhaustive (hybrid sensors — sharp fovea + low-acuity periphery, depth-only — might produce intermediate geometries).

Table 2: Foveation vs. uniform at $\sigma_{\max} = 8$ Gaussian blur with the same ResNet-18 encoder. Same on H1, different on H2 / behavioural / generalisation. Sources: Table 1, Figure 3, Figure 4, Figure 2c. Data are deterministic-rollout single-seed.

Measure	Foveated (fix)	Uniform	Verdict
LSTM GPS R^2 (Gibson)	$+0.06 \pm 0.88$	-0.31 ± 0.86	both chance, <i>indistinguishable</i>
Encoder $\rightarrow$ GPS R^2	-0.65 ± 0.28	-0.32 ± 0.08	both negative, overlapping
LSTM compass R^2 (Gibson)	$+0.07 \pm 0.69$	$+0.36 \pm 0.23$	uniform $\sim 5\times$ better
LSTM GPS R^2 (MP3D, held out)	$+0.35 \pm 0.27$	-0.36 ± 1.38	differ by 0.71
Shortcut SPL drop %	21%	41%	uniform $2\times$ more brittle
Memory-transplant cost (fov $\rightarrow$ uni)	-0.10 SPL beyond self-noise		not interchangeable
Pairwise CKA(fov, uni)	$< 10^{-4}$		near-orthogonal subspaces
1-NN purity (pooled)	1.000 vs. chance 0.20		linearly separable

log-polar matches uniform on H1, the encoder-spatial-output mechanism (§4.2, Proposed mechanism) would need reframing.¹¹

4.5 Probe-readable vs. policy-used memory: a 2×2 dissociation

H1 measures what is *readable* from the recurrent state; the shortcut-discovery experiment measures what the policy actually *relies on*. Paired same-scene-different-goal episodes (20 scenes $\times$ 10 pairs) compare second-episode SPL under reset vs. persistent LSTM: the SPL drop indexes how tightly the policy couples actions to integrated recurrent memory. Plotting it against probe-readable GPS (Figure 4) reveals two off-diagonal points: coarse has a linear top-layer GPS code that its policy weights below what the probe predicts, and uniform has no linear GPS **yet relies heavily on memory** — **presumably via non-spatial scene-history features the linear-GPS probe is blind to**. We label these candidate dissociations rather than established anomalies.¹²

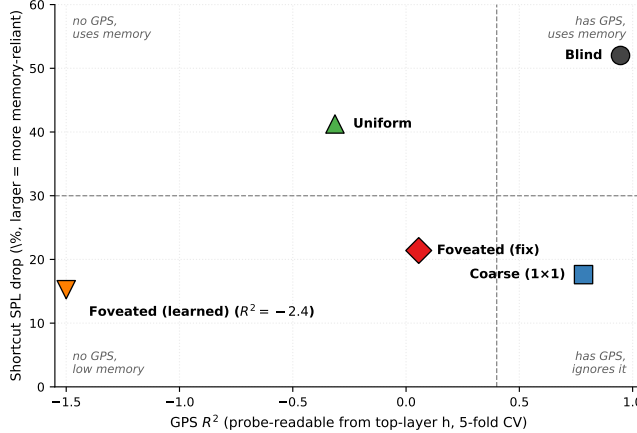


Figure 4: Probe-readable GPS (H1) vs. behavioural memory reliance (shortcut). The five conditions occupy four corners of a 2×2 : blind “has GPS, uses memory” (consistent with H1); foveated (fix) and foveated (learned) “no linear GPS, low memory reliance” (also consistent with H1); **coarse** and **uniform** are the two off-diagonal candidate dissociations (single-seed; multi-seed gates the “anomaly” label). Foveated (learned)’s $R^2 = -2.43$ is clipped at -1.5 .

A trajectory-level look at *how* persistent-memory failures manifest is consistent with the dissociation (Figure 5). Aggregating across all same-floor paired-episode failures (Figure 5, right panel): uniform is the only condition with a meaningful sample size whose persistent agent ends up closer to the

¹¹The foveation-vs-uniform contrasts in Table 2 are single-seed; multi-seed replication is in flight. F3 log-polar is currently $\sim 14\text{M} / 250\text{M}$ frames at submission.

¹²The 5 points are single-seed; confirming that “probe-readable” and “policy-relied-on” truly dissociate (rather than the off-diagonal points sitting within seed-to-seed noise) requires the multi-seed replication in flight.

previous-episode goal than the new one (margin +1.83m on $n=46$); blind / coarse / foveated all end nearer the new goal but never reach it.¹³ If the uniform pattern survives multi-seed replication, it would be consistent with the 2×2 : uniform’s memory would plausibly act as a visual / scene-history anchor that locks the agent in place, whereas blind’s GPS code would interfere via a different channel — the recurrent state mis-reporting current position to the policy’s goal-direction integrator.

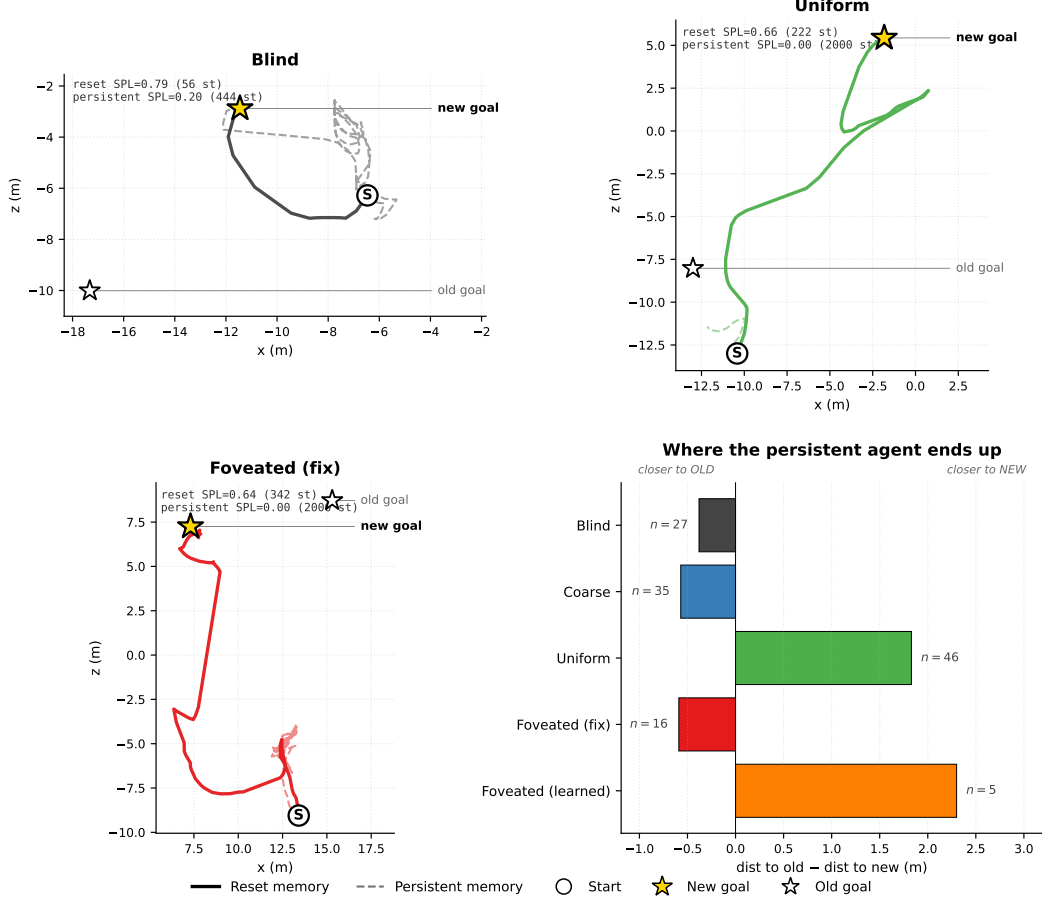


Figure 5: *Persistent-memory failure modes (one paired-episode case per condition + aggregate)*. The first three panels show *Blind* (persistent agent attempts the new goal, loops, gives up at 443 steps), *Uniform* (persistent agent stays at the old-goal location for the full 2000-step cap), and *Foveated* (persistent agent loops chaotically, reaches neither goal). Right panel: per-condition margin = (avg distance to old goal) – (avg distance to new goal) over all same-floor failures (reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30 , $|\Delta y_{\text{goal}}| < 0.5\text{m}$). Uniform shows a positive margin at meaningful $n=46$ (closer to the old goal); foveated (learned) shows the same direction at +2.30m but $n=5$ is too small to interpret. Single-seed; multi-seed replication is in flight.

4.6 Gaze location (H3, in progress)

H3 asks whether gaze *location* acts as a second representational-content axis within the rich-encoder regime. Our minimal learned-gaze module (deterministic sigmoid MLP, slow updates per 128-step segment) freezes at (0.49, 0.62) with per-dimension std $\leq 2.5 \times 10^{-4}$ after 250M frames; anti-collapse regularisers restore variance only at task-cost (Appendix E). The collapse is not evidence about active gaze in general — the decoder is minimal by design — but it provides an empirical anchor for the gaze-location test: a foveated-shifted control with gaze hardcoded at (0.49, 0.62)

¹³Foveated (learned) shares uniform’s positive-margin direction at +2.30m but $n=5$ is too small to interpret. The locking interpretation is single-seed and not yet established; multi-seed replication is in flight.

instead of $(0.5, 0.5)$, identical to foveated (fix) in every other respect, isolates static gaze location from the optimisation-path interaction that produced the collapsed gaze. Training is in progress at submission.

4.7 Boundaries

What does *not* discriminate also defines what our H1/H2 claims are about. Three null results bracket our findings. (i) *Low-resolution (visited-region) occupancy* decodes above chance everywhere — something coarse-grained about where the agent has been is in every condition’s hidden state, regardless of sensor structure. (ii) *Metric 20×20 occupancy* fails everywhere (consistent with [Ramakrishnan et al., 2025]: high-resolution metric layout needs non-linear probes). (iii) *The ego-frame goal vector* is at chance everywhere, because PointGoal supplies it as a per-step input — the policy has no incentive to re-encode it. Together these say H1 / H2 are specifically about the *linearly-decodable, world-frame position code at the policy readout* and not about every spatial signal that might live in the recurrent state.

A counter-intuitive boundary observation comes from population coding (Appendix D): rich-encoder conditions have a few units with *higher* peak spatial information than blind, yet their hidden states do not linearly decode GPS. *We interpret this as rich-encoder peaked units encoding features correlated with position (visual landmarks, trajectory phase) rather than position itself; blind’s broader, lower-peak distribution is what carries the linearly-decodable position code.* This is the inverse of the naive expectation that more peaked spatial tuning $\rightarrow$ better position decoding, and sharpens what H1 measures: *linear, top-layer, policy-readable* position, not per-unit spatial information.

5 Discussion

5.1 Two orthogonal axes: sensor structure and gaze location

The three findings cohere along two axes. *Magnitude axis (H1)*: how much of the policy’s spatial competence is carried by integrated recurrent memory varies inversely with what the encoder hands the LSTM per step. *Format axis (H2)*: even within the rich-encoder regime where the magnitude axis collapses to “not via linear top-layer GPS”, the recurrent state still carries condition-specific content that is not interchangeable across conditions — something is being learned per condition, but in linearly disjoint subspaces. *The 2×2 dissociation in §4.5 is what the two axes look like together: coarse has the H1 magnitude (linear GPS code) but its policy under-weights it; uniform lacks the H1 magnitude but the H2 format axis still carries a memory the policy heavily uses.*

Read this way, the encoder–memory race (H1) is a single dimension of a larger picture: *what the recurrent state carries is shaped by what the encoder doesn’t, and the format of that carrying is condition-specific even when the magnitude saturates.* We do not claim the underlying mechanism is the same in coarse (where the encoder is structurally bottlenecked) and uniform (where the encoder is rich but happens to discard linear GPS); they sit at different points on both axes. H3 (§4.6) probes whether gaze location is a third axis within the rich-encoder regime, perpendicular to both magnitude and format; the foveated-shifted control is in training.¹⁴

5.2 Biological precedent: sensor structure shapes hippocampal representation

The principle that sensor structure constrains spatial-memory format is not one we propose; biological evidence for it predates ours by decades. Our contribution is to isolate the principle in a controlled silicon model where the encoder, memory, policy, training distribution and reward are held fixed and only the visual sensor varies. Three threads of animal-navigation research are particularly close to the three findings.

H1 (sensor bottleneck $\rightarrow$ memory takes over). Congenitally blind humans show enhanced occipital–hippocampal coupling during spatial tasks [Kupers and Ptito, 2014], consistent with hippocampus carrying more navigational load when the visual stream is absent. Rodent grid-cell periodicity collapses under prolonged darkness while the animal still navigates [Chen et al., 2016], indicating

¹⁴4 coarse-recipient transplant cells are still pending; the asymmetry-direction along the magnitude axis is robust to landed cells, but completing the matrix is needed before the asymmetry-statement is exhaustive. The unifying account is a candidate; a direct causal test (visual ablation mid-rollout) is left for follow-up.

spatial behaviour can be sustained when the visual landmark stream is removed. Both are biological analogs of our blind / coarse regime, where the LSTM — deprived of rich visual variety per step — retains a linear top-layer GPS code that is no longer linearly readable in the rich-encoder conditions.

H2 (different sensors → different spatial codes). The primate vs. rodent contrast in hippocampal cell types — view-cells dominant in primates, place-cells dominant in rodents — has been attributed in part to the foveated-with-saccades vs. near-panoramic visual sampling regime each species uses [Rolls, 2024]. Cross-modality remapping in echolocating bats, where the same neurons recode under vision vs. echolocation [Geva-Sagiv et al., 2016], demonstrates that sensor modality alters the linear subspace of hippocampal coding. Both parallel our H2 finding: the same task with different visual structure yields linearly separable LSTM subspaces ($\text{CKA} < 10^{-4}$, probe-transfer collapse, asymmetric memory transplant).

H3 (gaze location modulates memory encoding). In primates, hippocampal theta is gated by saccades, and the reliability of this saccade-gated signal predicts memory encoding success [Jutras et al., 2013, Tas et al., 2016]. Where the eye points is therefore not memory-neutral. Whether the analogous prediction holds for an LSTM gaze module is what §4.6 is testing (foveated-shifted causal control in training).

We treat these as convergent (not conclusive) evidence. Convergence across systems — and across silicon vs. neural — is suggestive of a sensor–memory coupling principle more general than any single architecture, but it does not show that the mechanisms are the same. Our experiments isolate the principle in a controlled, single-architecture setting; biological work establishes that the principle exists in animals at all. The two are complementary; neither is decisive alone.

5.3 Why this, not alternative accounts

Four alternative explanations are ruled out: skill difference (success bunched 93–99%), probe capacity (DtG decodes everywhere), sample-size or stochastic-sampling artefacts (Appendix A), and linear-probe artefact (memory transplant reproduces the ordering outside the probe framework; MLP probe shows the divergence is specific to *linear* top-layer decodability, which is what the linear DD-PPO action head reads). Full rebuttals in Appendix C.

5.4 Implications and scope

The encoder–memory race is most fundamentally a claim about the *interface* between an upstream encoder and a downstream memory module — what the memory module carries depends on what the encoder doesn’t. Our LSTM-based PointNav agent is one tractable instantiation. **We expect the principle to apply, with adapted measurements, to any architecture that maintains a persistent state across timesteps and reads it via a linear or near-linear policy head:** transformer-based navigators (where “memory” lives in the KV cache or attention over history) should show analogous content-level dissociations when attended tokens are constrained the way our encoder bottleneck constrains spatial features. **The relevant axis is not pixel count: coarse’s 48^2 input with a 1×1 feature map has a strong linear top-layer GPS code while uniform’s 256^2 does not, so we attribute the pattern to encoder spatial-feature *variety* per step rather than raw input bandwidth.** Within recurrent agents specifically, encoder capacity becomes a practical training-time lever for inducing cognitive-map-like representations *if* the resolution scaling sweep (Appendix H) confirms causality. Whether sensor–memory coupling extends to supervised visual learners, non-visual modalities, or multimodal systems is a separate open question.

5.5 Limitations

(i) *Single seed* for every cross-condition number reported in this paper (H1 R^2 values, LSTM gain numbers, 5×5 transplant cells, shortcut SPL drops, lock-onto-old margins, MP3D cross-dataset shifts); multi-seed replication is in flight and gates the strength of every quantitative claim. (ii) *Foveated (fix) checkpoint* is ckpt.36 ($\sim 174\text{M}$ frames, last clean before a silent NaN-gradient corruption; Appendix F); the other four conditions use the final converged checkpoint. (iii) *Learned-gaze module* is a minimal deterministic sigmoid MLP that collapsed under pure task supervision; the H3 test (foveated-shifted control) does not depend on this module working. Richer gaze mechanisms (stochastic gaze, intrinsic-reward gaze, attention-based selection) are out of scope. (iv) *Foveation*

model: $\sigma_{\max} = 8$ Gaussian blur is a simplification; a log-polar control and an $\sigma_{\max} \in \{2, 4, 12, 20\}$ strength sweep are in training (Appendix G). (v) *Linear probes only*: non-linear probes might reveal structure Ridge misses (a 2-layer MLP partially recovers GPS in rich-encoder conditions; Appendix B). (vi) 5×5 *transplant matrix* has 4 unmeasured coarse-recipient cells (Figure 3, right); the asymmetry statement uses non-coarse-recipient cells only. (vii) *Architecture scope*: we use a recurrent (LSTM) backbone as a single-architecture case study for direct comparability with the emergent-maps literature [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018]. The encoder–memory race is a claim about an upstream-encoder $\rightarrow$ downstream-memory interface and should in principle apply to other architectures (transformer-based navigators, attention over history); we do not verify this here.

6 Conclusion

Using a recurrent (LSTM) PointNav agent as a controlled case study, a five-condition visual-pathway ablation suggests sensor structure as a representational-content axis: the encoder–memory race (H1), condition-specific representational format (H2), and a candidate probe-readable-vs.-policy-used dissociation at the $H1 \times \text{shortcut}$ intersection. The H1 ordering reproduces on held-out MP3D, and the pattern parallels independent animal-navigation findings on hippocampal compensation under sensory deprivation and cross-modality remapping. The cross-condition observations are single-seed; multi-seed replication is in flight and gates the strength of every quantitative claim. Pending experiments that would tighten or falsify the account: a finer-grained encoder-resolution scaling sweep (Appendix H) for causal H1; the foveated-shifted control for H3; the four coarse-recipient transplant cells; the foveation strength sweep and log-polar control. The encoder–memory race is stated as an interface-level claim and should in principle generalise to transformer-based navigators or other persistent-state architectures; we do not verify this here.

References

- Andrei Atanov, Yicheng Fu, Gaurav Singh, Jianren Yu, Andrew Spielberg, and Amir Zamir. How far can a 1-pixel camera go? solving vision tasks using photoreceptors and computationally designed visual morphology. In *European Conference on Computer Vision (ECCV)*, 2024.
- Andrea Banino, Caswell Barry, Benigno Uria, Charles Blundell, Timothy Lillicrap, Piotr Mirowski, Alexander Pritzel, Martin J Chadwick, Thomas Degris, Joseph Modayil, et al. Vector-based navigation using grid-like representations in artificial agents. *Nature*, 557(7705):429–433, 2018.
- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.
- Guifen Chen, John A. King, Neil Burgess, and John O’Keefe. Absence of visual input results in the disruption of grid cell firing in the mouse. *Current Biology*, 26(17):2335–2342, 2016.
- Christopher J Cueva and Xue-Xin Wei. Emergence of grid-like representations by training recurrent neural networks to perform spatial localization. In *International Conference on Learning Representations (ICLR)*, 2018.
- Kshitij Dwivedi, Gemma Roig, Aniruddha Kembhavi, and Roozbeh Mottaghi. What do navigation agents learn about their environment? In *Computer Vision and Pattern Recognition (CVPR)*, 2022.
- Russell A. Epstein, Eva Zita Patai, Joshua B. Julian, and Hugo J. Spiers. The cognitive map in humans: spatial navigation and beyond. *Nature Neuroscience*, 20(11):1504–1513, 2017.
- Maya Geva-Sagiv, Sandro Romani, Liora Las, and Nachum Ulanovsky. Hippocampal global remapping for different sensory modalities in flying bats. *Nature Neuroscience*, 19(7):952–958, 2016.
- James Gornet and Matt Thomson. Automated construction of cognitive maps with visual predictive coding. *Nature Machine Intelligence*, 6:820–833, 2024.
- Jay A Hennig, Sandra A Romero Pinto, Takahiro Yamaguchi, Scott W Linderman, Naoshige Uchida, and Samuel J Gershman. Emergence of belief-like representations through reinforcement learning. *PLOS Computational Biology*, 19(9):e1011067, 2023.

- John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2019.
- Michael J. Jutras, Pascal Fries, and Elizabeth A. Buffalo. Oscillatory activity in the monkey hippocampus during visual exploration and memory formation. *Proceedings of the National Academy of Sciences*, 110(32):13144–13149, 2013.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning (ICML)*, 2019.
- Ron Kupers and Maurice Ptito. Compensatory plasticity and cross-modal reorganization following early visual deprivation. *Neuroscience & Biobehavioral Reviews*, 41:36–52, 2014.
- Volodymyr Mnih, Nicolas Heess, Alex Graves, and Koray Kavukcuoglu. Recurrent models of visual attention. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2014.
- Motahareh Pourrahimi and Pouya Bashivan. Emergent brain-like representations in a goal-directed neural network model of visual search. *bioRxiv preprint*, 2025.
- Santhosh Kumar Ramakrishnan, Erik Wijmans, Philipp Krähenbühl, and Vladlen Koltun. Does spatial cognition emerge in frontier models? In *International Conference on Learning Representations (ICLR)*, 2025.
- Edmund T. Rolls. Why do primates have view cells instead of place cells? *Trends in Cognitive Sciences*, 28(9):862–874, 2024.
- Ruth Rosenholtz. Capabilities and limitations of peripheral vision. *Annual Review of Vision Science*, 2:437–457, 2016.
- Manolis Savva, Abhishek Kadian, Oleksandr Maksymets, Yili Zhao, Erik Wijmans, Bhavana Jain, Julian Straub, Jia Liu, Vladlen Koltun, Jitendra Malik, Dhruv Batra, and Devi Parikh. Habitat: A platform for embodied AI research. In *International Conference on Computer Vision (ICCV)*, 2019.
- Jinghuan Shang and Michael S. Ryoo. Active vision reinforcement learning under limited visual observability. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Nathan Sprague and Dana H. Ballard. Eye movements for reward maximization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2003.
- Hans Strasburger, Ingo Rentschler, and Martin Jüttner. Peripheral vision and pattern recognition: A review. *Journal of Vision*, 11(5):13, 2011.
- Amanda C. Tas, Steven J. Luck, and Andrew Hollingworth. The role of attention in the encoding of visual working memory. *Journal of Experimental Psychology: Human Perception and Performance*, 42(8):1121–1138, 2016.
- Erik Wijmans, Abhishek Kadian, Ari Morcos, Stefan Lee, Irfan Essa, Devi Parikh, Dhruv Batra, and Manolis Savva. DD-PPO: Learning near-perfect pointgoal navigators from 2.5 billion frames. In *International Conference on Learning Representations (ICLR)*, 2020.
- Erik Wijmans, Manolis Savva, Irfan Essa, Stefan Lee, Ari Morcos, and Dhruv Batra. Emergence of maps in the memories of blind navigation agents. In *International Conference on Learning Representations (ICLR)*, 2023.
- Sylvia Wirth, Pierre Baraduc, Angélique Planté, Serge Pinede, and Jean-René Duhamel. Gaze-informed, task-situated representation of space in primate hippocampus during virtual navigation. *PLOS Biology*, 15(2):e2001045, 2017.
- Fei Xia, Amir R Zamir, Zhiyang He, Alexander Sax, Jitendra Malik, and Silvio Savarese. Gibson Env: Real-world perception for embodied agents. In *Computer Vision and Pattern Recognition (CVPR)*, 2018.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive” or “we were unable to find the license for the dataset we used”). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading “NeurIPS Paper Checklist”.**
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers.**

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer:
answerYes

Justification: The abstract and §1 state the three hypotheses H1/H2/H3 with explicit signatures (what pattern supports each), and preview the three findings with specific magnitudes (path-history $R^2=0.56$, CKA <0.004 , compass $R^2=0.94$ vs GPS $R^2=0.03$). These are expanded in §4 and §5 with no stronger claims than the data support.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer:

answerYes

Justification: §5.4 explicitly lists six limitations (i–vi) covering checkpoint staleness, single-seed H3, missing proposed analyses, foveation-model simplification, linear-probes-only, and LSTM-architecture scope.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer:

answerNA

Justification: The paper reports empirical results; it contains no theoretical claims or proofs.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer:
answerYes

Justification: §3.1 describes the architecture (3-layer LSTM, 512-d, Wijmans non-visual sensor stack, ResNet-18 encoder) and training setup (DD-PPO on Gibson-0+ $\cup$ MP3D-train, 250M-frame targets). §3.2–§3.6 describe all probing and cross-condition protocols. Appendix A details the gaze-diversity ablation. Code will be released upon publication.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer:
answerYes

Justification: §3.7 commits to open-source release of the training and probing code. Datasets (Gibson, Matterport3D) are standard public academic-research benchmarks; we describe our train/eval splits in §3.1.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).

- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer:

answerYes

Justification: §3.1 specifies training pool and eval set; §3.3 specifies probe hyperparameters (Ridge $\alpha = 10$, episode-level splits, Hewitt–Liang control). Appendix A gives the ablation’s loss formulation and coefficient.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer:

answerNo

Justification: We report single-seed numbers per condition. This is documented as Limitation (ii) in §5.4. Multi-seed replication for tighter confidence intervals is in the immediate follow-up.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).

- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer:

answerYes

Justification: §3.7 reports per-condition training budget (5–7 days on 2×V100 for 250M frames) and probing budget (<1h per condition on one V100).

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer:

answerYes

Justification: The research uses public academic-research benchmarks, publicly-released open-source RL frameworks, and involves no human subjects or deployment. We also report and remediate a numerical-stability bug upstream (Appendix B).

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer:

answerYes

Justification: §5.3 discusses implications — sensor design as a tunable axis for interpretable embodied agents. Negative societal impacts are minimal: the work is fundamental research on simulated indoor navigation with no direct deployment pathway.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.

- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer:

answerNA

Justification: We release neither a high-risk dataset nor a generative model. Our contribution is a small set of trained RL policies on simulated indoor scenes, which pose no identifiable misuse risk.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer:

answerYes

Justification: Gibson Database and Matterport3D are used under their released academic-research licenses; Habitat-Sim and habitat-lab are MIT-licensed. Our use complies with their terms; citations §3.1.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.

- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer:

answerYes

Justification: Code will be released with documentation upon publication. Figures are novel; they are released under the paper's license alongside the code. The gaze-diversity auxiliary loss and NaN-sanitisation patch are small standalone modules with docstrings.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer:

answerNA

Justification: The paper involves no crowdsourcing and no human-subjects data.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer:

answerNA

Justification: No human-subjects research; no IRB/equivalent review was required.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer:

answerNA

Justification: LLMs were used only for writing assistance on the manuscript. They are not part of the core research methodology; all empirical results come from standard RL training and linear probing.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.

A Probing protocol details

Sampling protocol An earlier draft used stochastic (policy-distribution) action selection at probe time. Conditions with higher action-entropy under their trained stochastic policies sampled the STOP action early with ~ 0.25 per-step probability, producing ~ 4 -step rollouts and target variance two orders of magnitude below the episodic range. The resulting probe R^2 values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons; we switched to the deterministic (argmax) protocol used here and elsewhere in our pipeline. All paper results use the deterministic protocol.

Length-matching ablation. Under deterministic rollouts, episode lengths span ~ 100 – 400 mean steps depending on condition. We use 5-fold episode-level CV to keep splits balanced per scene rather than length-matching across conditions. **[TODO: A controlled length-matching ablation (truncate every condition to a common length) is left for follow-up; we expect it to leave the H1 ordering unchanged because the bottleneck conditions decode at high R^2 even at the smallest sub-sample, but have not verified.]**

Cross-heading probe. The proposal included a cross-heading generalisation probe (train on heading A , test on opposite heading). Our implementation returned “insufficient samples” sentinels at our probe sizes; not in our results.

B Probing details (Layer 0 sensor branch and MLP probe)

LSTM Layer 0 sensor branch. The non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar, prev-action) projects each sensor to 32-d, concatenates them with the visual encoder’s flattened output, and feeds the result as the LSTM’s Layer 0 input. The GPS embedding therefore enters the LSTM via this concatenation; this explains why Figure 2c shows linearly-decodable GPS at Layer 0 even for conditions whose visual encoder discards GPS entirely.

MLP probe details. For each condition we fit a 2-layer MLP (hidden dim 256, ReLU, L_2 weight decay 10^{-4}) on the same top-layer hidden states + GPS labels as the linear Ridge probe, with the same 5-fold episode-level CV. Top-layer recovery: foveated $R^2 = 0.51$, foveated (learned) 0.73, uniform 0.55 (vs. Ridge’s chance R^2). The MLP probe alone cannot disentangle “non-linear position code” from “MLP regressing through position-correlated features (wall distance, scene identifiers, episode-phase markers)”; the resolution scaling sweep (Appendix H) tests this more directly by varying input resolution at fixed encoder stack.

C Alternative-account rebuttals (full)

Task-competence differences. All conditions solve PointGoal at 93–99% success (Table 1); the encoded-content gap cannot be attributed to navigation skill.

Probe-capacity / probe-fitting artefacts. DtG decodes robustly across all five conditions ($R^2 \geq 0.81$), demonstrating Ridge has the capacity to fit high- R^2 scalars when the hidden state contains them. Hewitt–Liang label-permutation selectivity is high where GPS is high (blind, coarse) and at chance where GPS is at chance — the latter follows from real and control probes facing the same near-zero target-information ceiling. The chance-level rich-encoder R^2 is therefore a hidden-state property, not a probe artefact.

Sample-size / trajectory-distribution artefacts. CV σ spans come from the same 500-episode pool with ~ 100 – 400 -step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft is documented and retired in Appendix A.

Linear-probe artefacts. Two complementary checks. (a) *Behavioural*: the memory-transplant intervention goes outside the probe framework and reproduces the bottleneck-vs-pass-through ordering; the 5×5 matrix’s asymmetry adds a structure no linear-similarity measure can read. (b) *Non-linear probe*: a 2-layer MLP recovers $R^2 \in [0.51, 0.73]$ from rich-encoder top-layer states (Appendix B). The bottleneck-vs-pass-through divergence is therefore not about whether *any* probe can recover position; it is specifically about whether a *linear* top-layer probe can. Since the DD-PPO action head is a linear projection from $\mathbf{h}_2$, the linear metric is what the policy itself can use, so H1 maps onto policy-accessible encoding regardless of how the MLP interpretation resolves.

D Supplementary visualisations

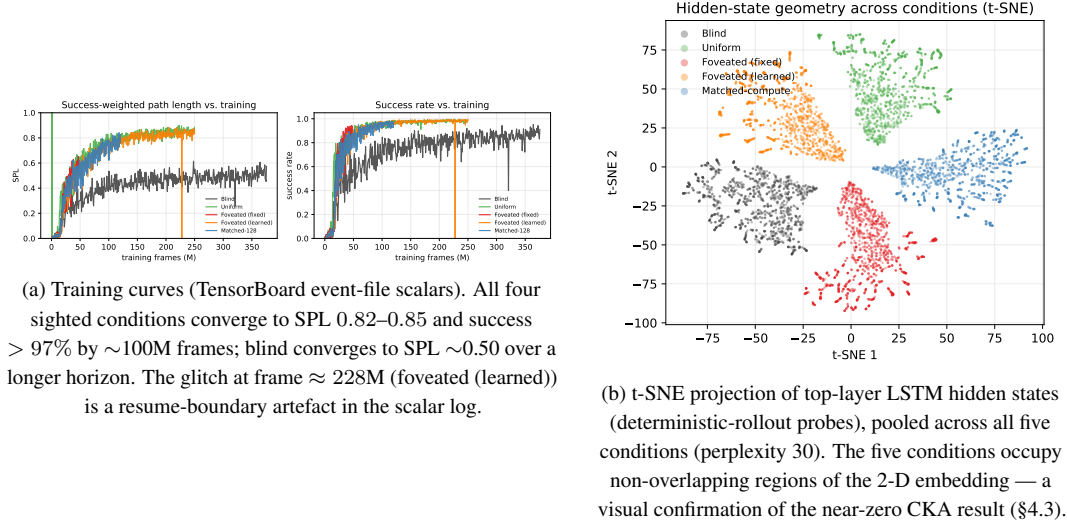


Figure 6: Supplementary visualisations referenced from the main text.

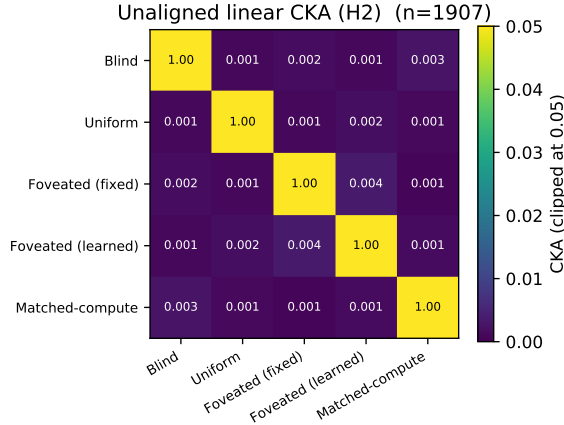


Figure 7: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states ($n=95,640$ steps per condition under deterministic-rollout probes); every off-diagonal is below 10^{-4} . Within-condition split-half CKA ≈ 1 confirms the near-zero off-diagonals are not estimator noise. Pairs that look identical here differ markedly under transplant and probe-transfer (Figure 3); CKA is the least informative of the three H2 measures because it cannot read the asymmetric structure that transplant and probe-transfer show.

E Gaze-Diversity Auxiliary Loss (H3 Ablation)

Motivation. In the first foveated (learned) training run the gaze decoder collapsed (per-dimension $\text{std} \approx 5 \times 10^{-4}$). We attribute this to two mechanisms: (i) once the sigmoid output saturates, its derivative $\sigma'(x) \rightarrow 0$ attenuates gradient signal; (ii) consistent-gaze views yield more predictable visual features, so the reward-gradient path has no explicit incentive for variety.

Target-variance hinge. We add $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma_{\text{target}}^2 - \text{Var}_{\text{batch}}[g])$ where $g \in [0, 1]^2$, $\text{Var}_{\text{batch}}$ averaged over the two gaze dimensions, $\sigma_{\text{target}}^2 = 0.01$ ($\text{std} \approx 0.1$), $\lambda = 0.01$. The hinge disappears once variance exceeds the target. We also evaluated an un-capped $-\lambda \cdot \text{Var}[g]$ variant as a negative control.

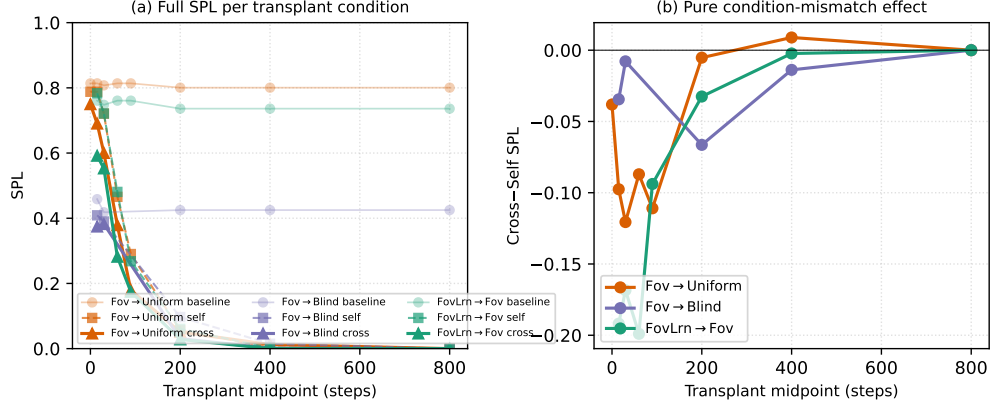


Figure 8: Memory-transplant midpoint sweep on three representative donor→recipient pairs (foveated→uniform, foveated→blind, foveated (learned)→foveated) across midpoints $\{0, 15, 30, 60, 90, 200, 400, 800\}$ time-steps ($n=100\text{--}150$ episodes/cell). The full 5×5 matrix in Figure 3 (right) is at midpoint $t=30$. (a): full SPL per condition — baseline / self / cross. The $t=0$ self/cross collapse to a protocol-noise floor (so the diagonal in Figure 3 is correctly read as rollout-divergence noise, not condition mismatch); SPL decays for all conditions at $t \geq 200$ because by then most successful episodes have already terminated, leaving only long atypical episodes. (b): cross-minus-self isolates condition-mismatch alone. The gap peaks in the $t \in [30, 90]$ window — where the agent has integrated history but is not yet committed to a trajectory — and decays toward 0 at $t \geq 400$. The asymmetry pattern reported at $t=30$ in Figure 3 is therefore neither a $t=30$ -specific artefact nor a late-midpoint inflation.

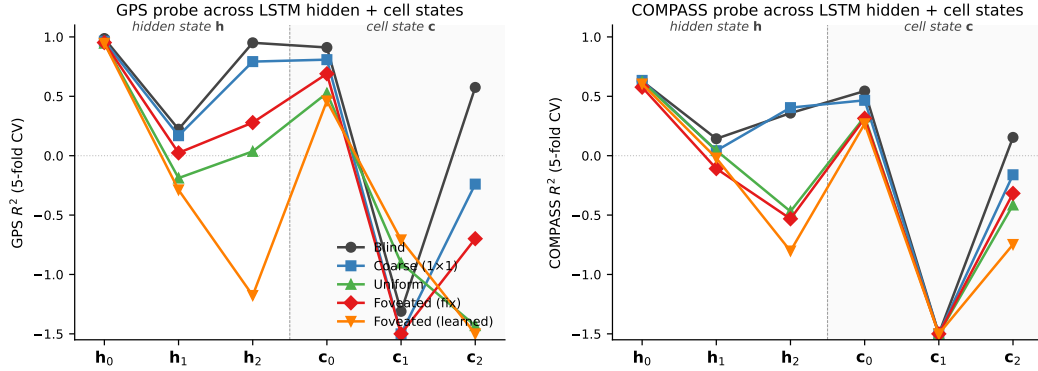


Figure 9: Linear probe (R^2 , 5-fold CV) at each LSTM hidden state h_ℓ and cell state c_ℓ , $\ell \in \{0, 1, 2\}$, for all 5 conditions. *Left: GPS. Right: compass.* Three patterns. (i) At h_0 all conditions decode GPS at $R^2 \sim 0.95$ (the GPS sensor embedding enters the LSTM concat there, regardless of vision condition). (ii) h_1 dips for every condition including blind ($R^2 = +0.22$); information passes through middle layer with reduced linear decodability for everyone, then partly recovers at h_2 for bottleneck. (iii) Cell states c_ℓ do not carry the GPS code as cleanly as h_ℓ : c_1 is at chance/negative for every condition; c_2 is positive only for blind ($R^2 = +0.57$). The H1 claim about “top-layer linearly-decodable GPS” is therefore specifically about h_2 (the policy readout), not the cell-state pathway. Same single-seed limitation as the main paper.

Pilot 1 (uncapped, job 2840256). After 10 M frames, gaze variance exceeded that of a uniform distribution over $[0, 1]^2$ (std $[0.43, 0.42]$ vs. theoretical max 0.289): gaze was essentially random. Task performance collapsed: SPL plateaued at 0.05, running-mean metrics transitioned to NaN at ~ 8 M frames. The uncapped regulariser destroyed the consistent-foveation signal the policy needs.

Pilot 2 (hinge, job 2842288). Gaze converged to an extreme corner of image space: mean $(0.001, 0.998)$ with std $[0.020, 0.023]$ — well below the target std of 0.1 and therefore with the

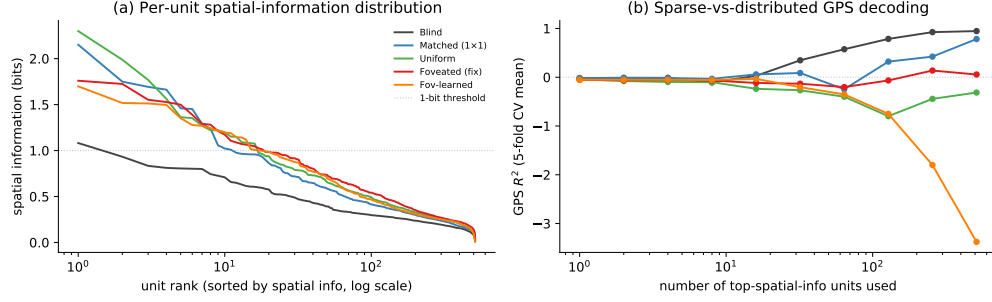


Figure 10: Population coding analysis (referenced from §4.7). (a) Per-unit spatial-information distribution: each curve is one condition’s units sorted by spatial information (descending), x-axis log scale. Counter-intuitively, rich-encoder conditions (uniform / foveated / foveated (learned)) have a few more peaked spatially-tuned units (max ≈ 2 bits) than bottleneck conditions; blind has the flattest distribution (max ≈ 1 bit, no unit reaches the dashed 1-bit threshold). (b) Sparse-vs-distributed GPS decoding: probe R^2 when only the top- k spatial-info units are used as features. Blind reaches $R^2 \approx 1.0$ at $k = 512$; coarse climbs to $R^2 \approx 0.5$; rich-encoder networks fail to decode GPS at any k even though their top units carry higher per-unit spatial information. The combined picture: *higher per-unit spatial information $\neq$ position readout*. Rich-encoder peaked units appear to encode features correlated with position (e.g. visual landmarks, trajectory phase) rather than position itself.

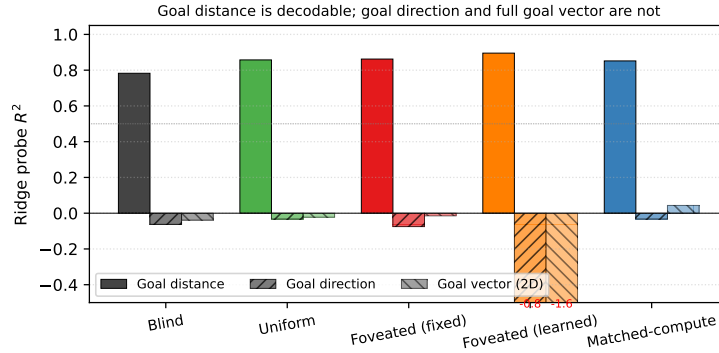


Figure 11: Goal-vector probe R^2 (referenced from §4.7). Goal *distance* (DtG) decodes strongly across all conditions ($R^2 \in [0.81, 0.90]$, validating probe machinery); goal *direction* and the full ego-frame goal vector are at chance or negative everywhere. PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it.

hinge active throughout. Task SPL plateaued at 0.05 as in pilot 1. The policy apparently found the minimum-task-cost gaze that partially satisfies the diversity pressure: a sigmoid saturated at one corner, producing a constant image-corner view.

Interpretation. Both pilots argue that in our deterministic sigmoid-MLP gaze architecture with slow-gaze rollouts, gaze collapse is a pre-requisite for task learning rather than an accident. Task signal alone does not spontaneously produce informative fixation, and an exogenous anti-collapse regulariser moves gaze toward whatever region reduces regulariser loss fastest, which is generically *not* the task-useful region. A genuine H3 test likely requires (i) a stochastic gaze location with entropy bonus, so diversity is sampled at rollout time rather than enforced on a deterministic output; or (ii) a task-aligned intrinsic reward (e.g., information gain about goal-relative position); or (iii) an architectural change separating gaze generation from visual-feature extraction (e.g., attention-based gaze).

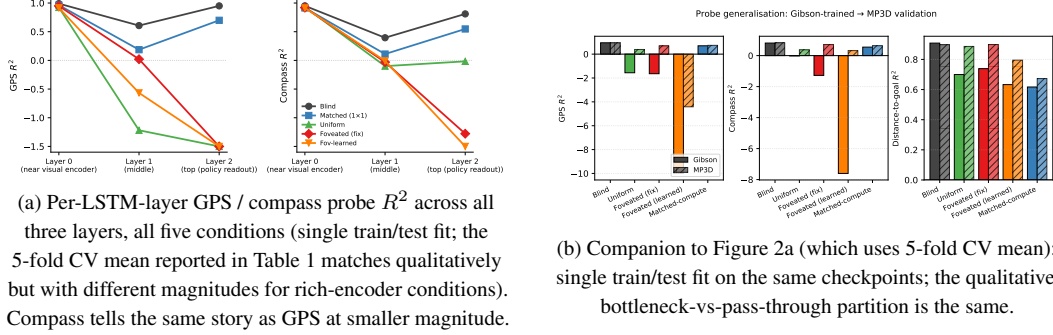


Figure 12: Per-layer and MP3D companion figures (referenced from §4.2).

F Training Stability: Silent Weight Corruption in DD-PPO

During an initial foveated training run we observed a failure mode that did not manifest as a crash but as slow, silent corruption of the learned weights. After $\approx 170\text{M}$ frames of stable progress (reward ≈ 10 , SPL ≈ 0.83), one mini-batch on one DDP worker produced a NaN gradient. Because `torch.nn.utils.clip_grad_norm_` does not sanitise non-finite gradients—the global norm becomes NaN, the clip coefficient becomes NaN, and `grad *= NaN` remains NaN—the subsequent `optimizer.step()` wrote NaN into every parameter. DDP all-reduce propagated the corruption across workers. The training loop did not raise: it continued for 2.5 days with reward=NaN and SPL=0 until `torch.multinomial` finally rejected the NaN probability tensor. All checkpoints saved during this window were unrecoverable (90 of 97 parameter tensors entirely NaN).

To isolate the origin of the NaN gradient we re-ran training from the last clean checkpoint with `torch.autograd.set_detect_anomaly(True)` and per-module forward-output hooks. The anomaly traceback pinned the failure to `MseLossBackward0`, i.e. the backward pass of the value-head loss $\frac{1}{2}(V(s) - R)^2$; rare return-vs-value discrepancies overflowed float32 in the squaring step, producing non-finite gradients downstream.

Our mitigation is a minimally invasive patch to `PPO.before_step` that zeroes any non-finite gradient element before clipping. A bad mini-batch becomes a no-op update rather than a catastrophic one. The patch is a no-op on clean training paths. Across the five conditions reported here, no NaN events were observed during full training after deployment, and all final checkpoints contain only finite weights. We reported the underlying issue upstream (habitat-lab issue #2226) with a minimal reproduction and a proposed native fix.

G Foveation experiments: status and design

This appendix details the four controlled foveation experiments referenced from §4.4. All four are submitted; numbers will populate §4.4 in revision as they land.

Table 3: Foveation experiment status (submission time).

Experiment	Tests	Status
F1–F4 strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$)	encoder–memory race as continuous lever	in training
F3 log-polar foveation	spatial-sampling vs. blur bottleneck	in training
Encoder feature-map probe (coarse / uniform / foveated)	where position information is held	landed; see §4.4
Foveated-shifted (gaze (0.49, 0.62))	gaze-location axis (links to H3)	in training

Foveation strength sweep (F1–F4). We train four additional foveated agents at $\sigma_{\max} \in \{2, 4, 12, 20\}$ alongside the existing $\sigma_{\max} = 8$, holding gaze location, encoder stack, training budget, and training pool fixed. Expected signature: at low $\sigma_{\max}$ the foveated agent should look like uniform; at high $\sigma_{\max}$ it should approach coarse. The R^2 vs. $\sigma_{\max}$ curve directly tests the encoder–memory race as a continuous lever rather than a binary one.

Spatial sampling vs. blur (F3 log-polar). Gaussian blur removes high-frequency content but preserves spatial layout: a $\sigma_{\max} = 8$ foveated image still feeds the encoder 256×256 pixels, so the ResNet-18 stack still produces an 8×8 spatial feature map. A log-polar foveation [Strasburger et al., 2011, Rosenholtz, 2016] resamples with a non-uniform retinal grid (e.g. 64×64): the encoder spatial output drops to $\sim 2 \times 2$, much closer to coarse’s 1×1 than to uniform’s 8×8 . This isolates the variable our sharpened H1 mechanism identifies as the trigger: *spatial-feature variety in the encoder output*. Falsifiable prediction (written before result available): log-polar should produce LSTM GPS $R^2 \geq 0.3$, between coarse’s $+0.78$ and uniform’s ≈ 0 , with corresponding behavioural shifts. If instead log-polar matches uniform, the encoder-spatial-output-dimensionality mechanism is wrong. F3 in training ($\sim 14\text{M} / 250\text{M}$ frames at submission, ETA 2–3 days).

Foveated-shifted (gaze location). Training a foveated agent with hardcoded static gaze at $(0.49, 0.62)$ matches foveated (learned)’s collapsed gaze location while removing the learned-gaze module from the optimisation path. If foveated-shifted does *not* reproduce foveated (learned)’s specific behaviours (Gibson–MP3D compass swing, transplant ranking), those behaviours can be attributed to the optimisation-path interaction rather than to gaze location alone. Links to H3 (§4.6).

Why disclose this rather than wait. Each experiment has a falsifiable signature; we disclose the designs openly so (i) the submission is honest about what foveation-specific evidence already exists vs. what is in flight, and (ii) the reader can identify which incoming result would change the paper’s interpretation.

H Encoder-capacity scaling sweep

[TODO: The scaling sweep trains coarse variants at input resolutions $\{32, 48, 64, 96, 128, 192\}$ alongside the existing blind (no encoder) and uniform 256×256 anchors. Probe results on the trained checkpoints will populate Figure 13 and this appendix section. Hypothesis: GPS probe R^2 decreases monotonically with input resolution, from $R^2 \approx 0.95$ at no-encoder to $R^2 \approx 0$ at 256×256 , crossing into the pass-through regime between 48 and 128 pixels.]



Figure 13: Encoder-capacity scaling curve. **[TODO: Pending data.]** x-axis: input resolution in pixels; y-axis: GPS and compass probe R^2 (5-fold CV, $\text{mean} \pm \sigma$).